وَمَا أُوتِيتُم مِّنَ الْعِلْمِ إِلَّا قَلِيلًا

Ain Shams University – Faculty of Engineering – ECE Dept. – Integrated Circuits Lab.
Dr. Hesham Omran

Analog IC Design (Xschem, Ngspice, ADT)
Lab 01
LPF Simulation and MOSFET Characteristics

Intended Learning Objectives

This lab is divided into two parts:

- In Part 1 you will
  - Get familiar with the design and simulation tools: Xschem and Ngspice.
  - Learn how to run transient and ac simulations.
  - Learn how to run parametric sweeps.
- In Part 2 you will
  - Learn the difference between dc sweep and parametric sweep.
  - Compare the behavior of PMOS and NMOS transistors.
  - Compare the behavior of short-channel and long-channel MOSFETs.
  - Get familiar with ADT.

Useful Resources

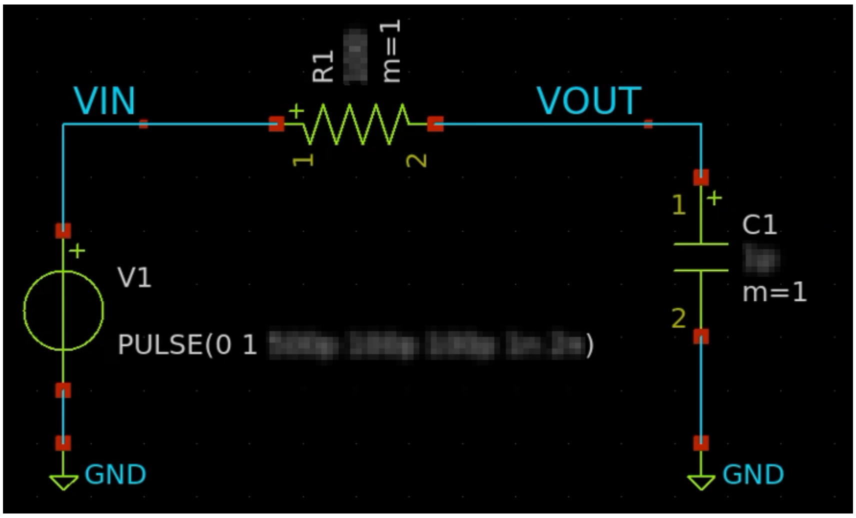
- ITI Analog IC Design Labs tutorial videos
- Xschem manual: http://repo.hu/projects/xschem/xschem_man/xschem_man.html
- Ngspice manual: https://ngspice.sourceforge.io/docs/ngspice-html-manual/manual.xhtml
- Useful xschem shortcuts:

Shortcut	Function
Ctrl+i	Insert instance
M	Move
Shift+M	Move while keeping wire connections (stretch)
C	Copy
Shift+R	Rotate
W	Start drawing wire from the current point
Q	Open properties window of the selected instance
Alt+L	Create label for a wire
Ctrl+A	Select all
F	Zoom fit
Ctrl+s	Save
Shift+T	Toggle if the selection is ignored or active in the spice simulation (grey is ignored and white is active)

PART 1: Low Pass Filter Simulation (LPF)

1. Transient Analysis

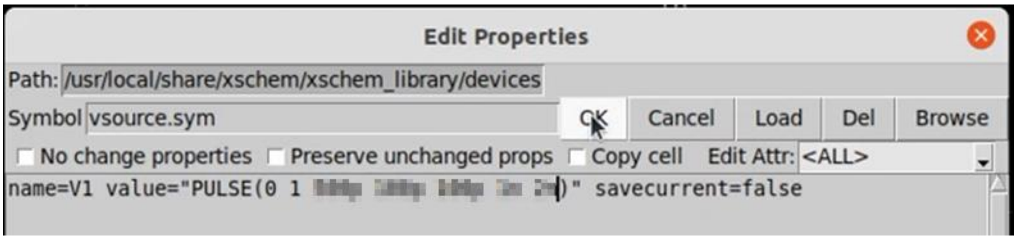
1) Create a new folder lab_01.
2) Open the terminal in this folder and start xschem.
3) Design a first order low pass filter that has $R = 1k\Omega$ and 1ns time constant.



4) Apply a square wave input with $V1 = 0, V2 = 1, Delay\ (TD) = 5ns, Pulse\ Width\ (PW) = 10ns, Period\ (PER) = 20ns$, and $T_{rise}\ (TR) = T_{fall}\ (TF) = \frac{PW}{100} = 100ps$.

➔ Hint: Ngspice pulse source format:

```
PULSE(V1 V2 TD TR TF PW PER NP)
```



5) Report transient analysis results for two periods (use max time step = $Period/100$).

➔ Hint: Ngspice command to run transient analysis (we don't use a dot before the command when it is in a control section):

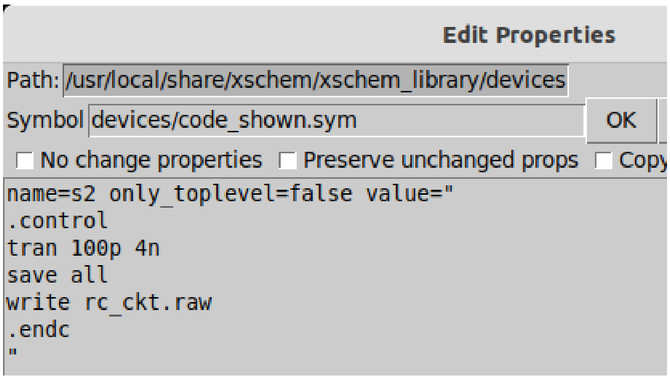
```
tran tstep tstop
```

➔ Note: We run spice in interactive mode and use a write command to explicitly write the output raw file.

➔ Ngspice control section example to run transient analysis.

```
.control
save all
tran 100p 4n
write rc_ckt.raw
```

```
.endc
```



6) Calculate rise and fall time (10% to 90%) using measure command. Export the results to an output text file.

➔ Hint: Measure command example (we don't use a dot before the command when it is in a control section). Note that we use "+" to write the command on multiple lines.

```
meas tran t_rise TRIG v(vout) VAL=0.1 RISE=1
+ TARG v(vout) VAL=0.9 RISE=1
```

➔ Ngspice control section example (modify it as needed for your simulation):

```
.control
save all
tran 100p 4n
write rc_ckt.raw
meas tran t_rise TRIG v(vout) VAL=0.1 RISE=1
+ TARG v(vout) VAL=0.9 RISE=1
print t_rise >> tran_result.txt
.endc
```

7) Derive an analytical expression (i.e., equation) for the rise and fall times as a function of the RC time constant. Compare simulation with analytical results in a table.
8) Do parametric sweep for $R = 1:1:5k\Omega$. Report overlaid results. Comment on the results.

➔ Hint: This is an example of a control section for transient analysis parametric sweep. Modify it as needed.

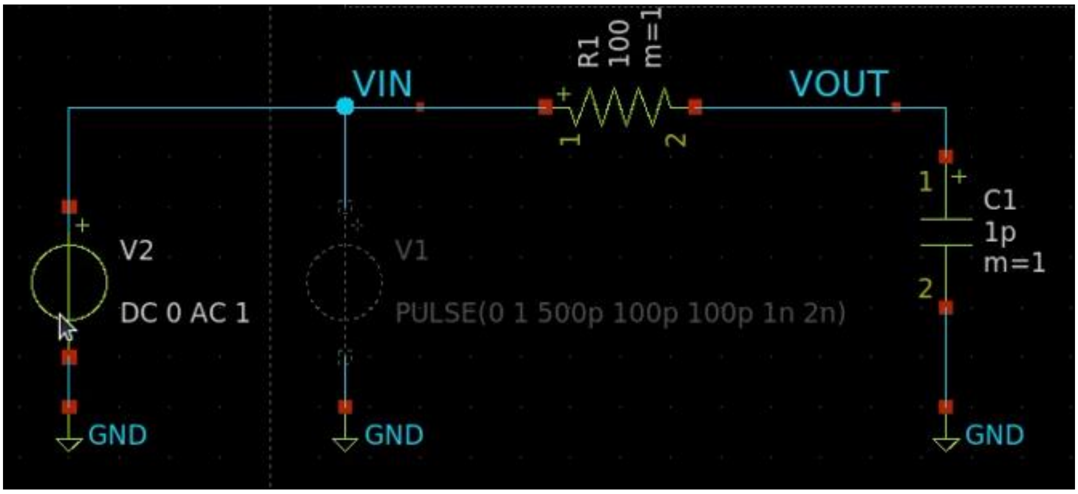
```
.control
save all
let R_val = 100
let R_stop = 300
let R_step = 100
while R_val le R_stop
  alter R1 R_val
  tran 100p 4n
  write rc_ckt.raw
  set appendwrite
  meas tran t_rise TRIG v(vout) VAL=0.1 RISE=1
  + TARG v(vout) VAL=0.9 RISE=1
```

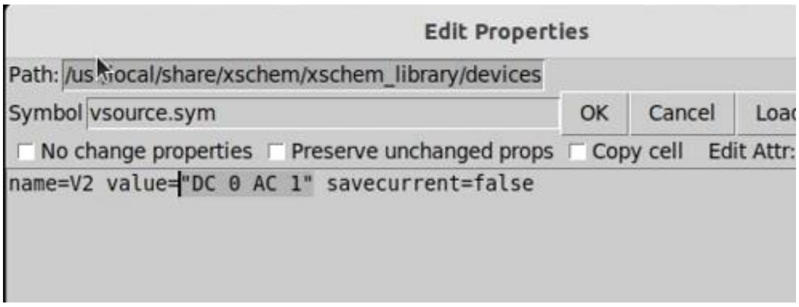
```
  print R_val t_rise >> tran_result.txt

  let R_val = R_val + R_step
end
.endc
```

2. AC Analysis

1) Modify your testbench to perform ac analysis.

➔ Hint: Use shift+T to make the transient source inactive in the schematic.





2) Report Bode Plot (magnitude and phase) for the previous LPF.
3) Calculate DC gain and 3dB bandwidth using measure command. Export the results to an output text file.

➔ This is an example of a control section for ac analysis. Modify it as needed.

```
.control
save all
ac dec 10 1 10g
write rc_ckt.raw
meas ac MAX_GAIN MAX vmag(vout) FROM=1 TO=10G
meas ac BW WHEN vmag(vout)=0.707 FALL=1
print MAX_GAIN BW >> ac_result.txt
.endc
```

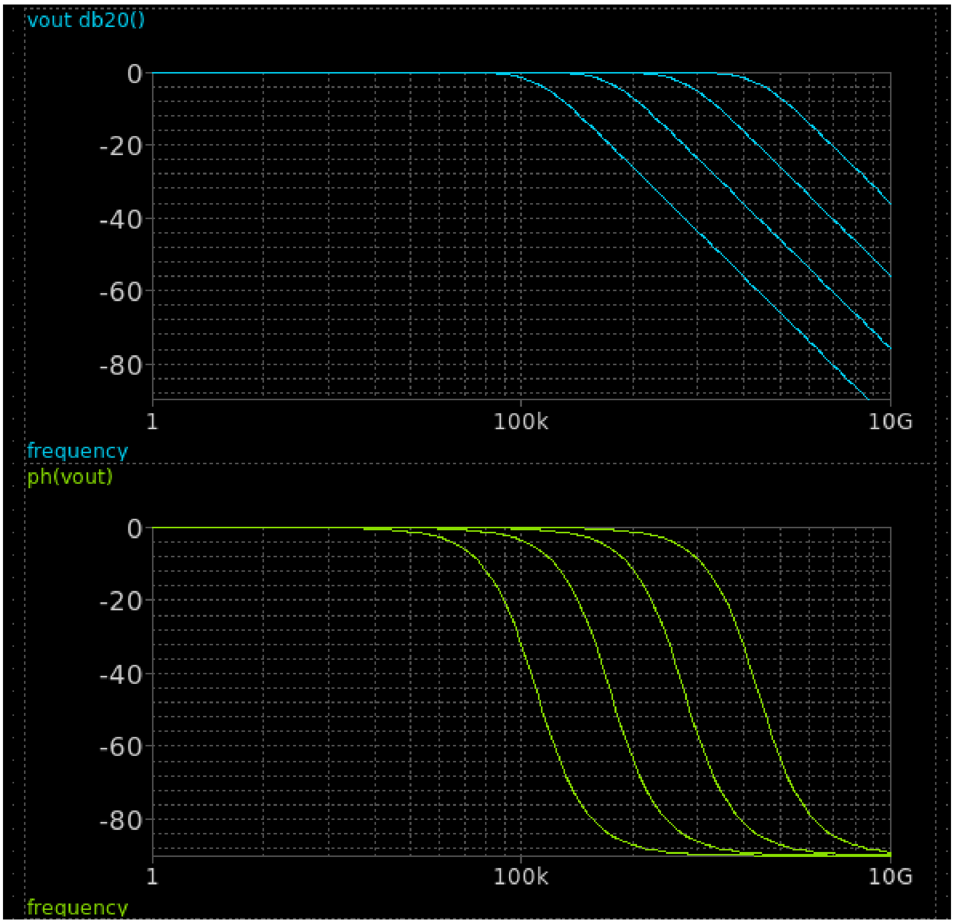
4) Derive an analytical expression (equation) for the bandwidth. Compare simulation with analytical results in a table.

5) Do parametric sweep for $R = 1,10,100,1000k\Omega$. Report overlaid results. Comment on the results.

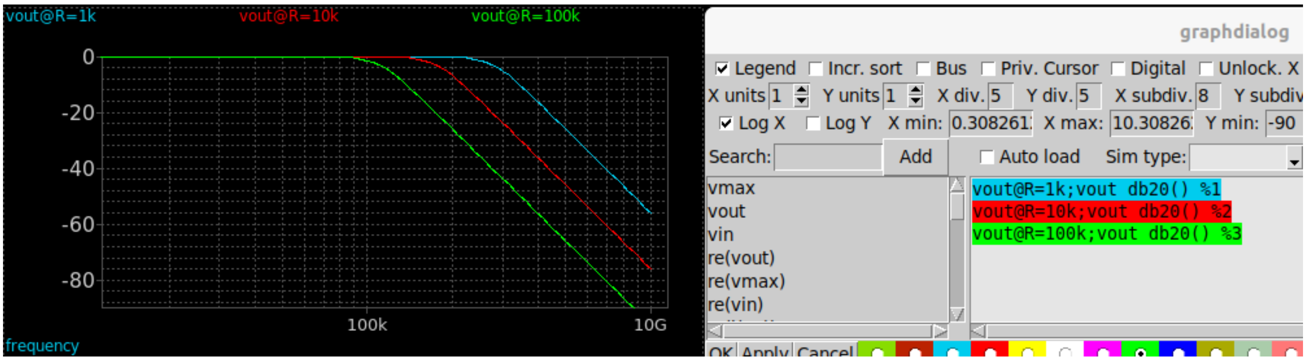
➔ This is an example of a control section for ac analysis parametric sweep. Modify it as needed.

```
.control
save all
let R_val = 1k
let R_stop = 1meg
let R_mult = 10

while R_val le R_stop
  alter R1 R_val
  ac dec 10 1 10g
  write rc_ckt.raw
  set appendwrite
  meas ac MAX_GAIN MAX vmag(vout) FROM=1 TO=10G
  meas ac BW WHEN vmag(vout)=0.707 FALL=1
  print R_val MAX_GAIN BW >> ac_result.txt
  let R_val = R_val * R_mult
end

.endc
```

➔ Hint: Bode plot (magnitude and phase) can be drawn using Xschem embedded waveform graphs.



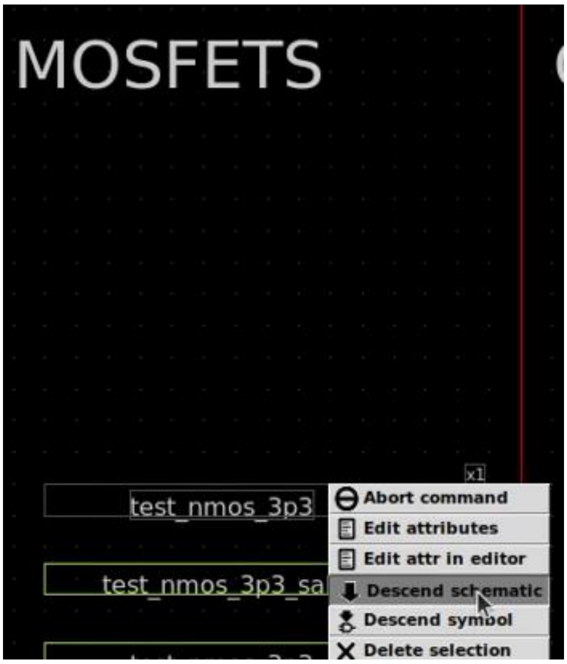
➔ Hint: You can define a different color and label for each trace in the parametric sweep as shown below.



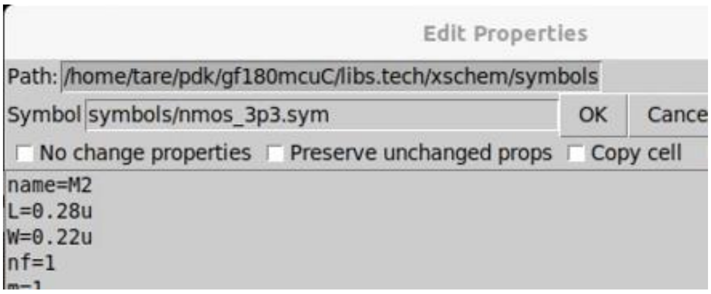
Part 2: MOSFET Characteristics

1. ID vs VGS (use Xschem)

1) Copy the GF180MCU xschemrc initialization file to your lab directory.
2) Run the export commands required for GF180MCU in your terminal or make sure they are automatically run in your ~/.bashrc file.
3) Open xschem and descend into the NMOS testbench.



4) **Do NOT overwrite the existing schematic. Use File -> Save as to save the schematic as a new schematic in your lab directory.** Note that the nmos properties should have the symbol path set as symbols/nmos_3p3.sym.



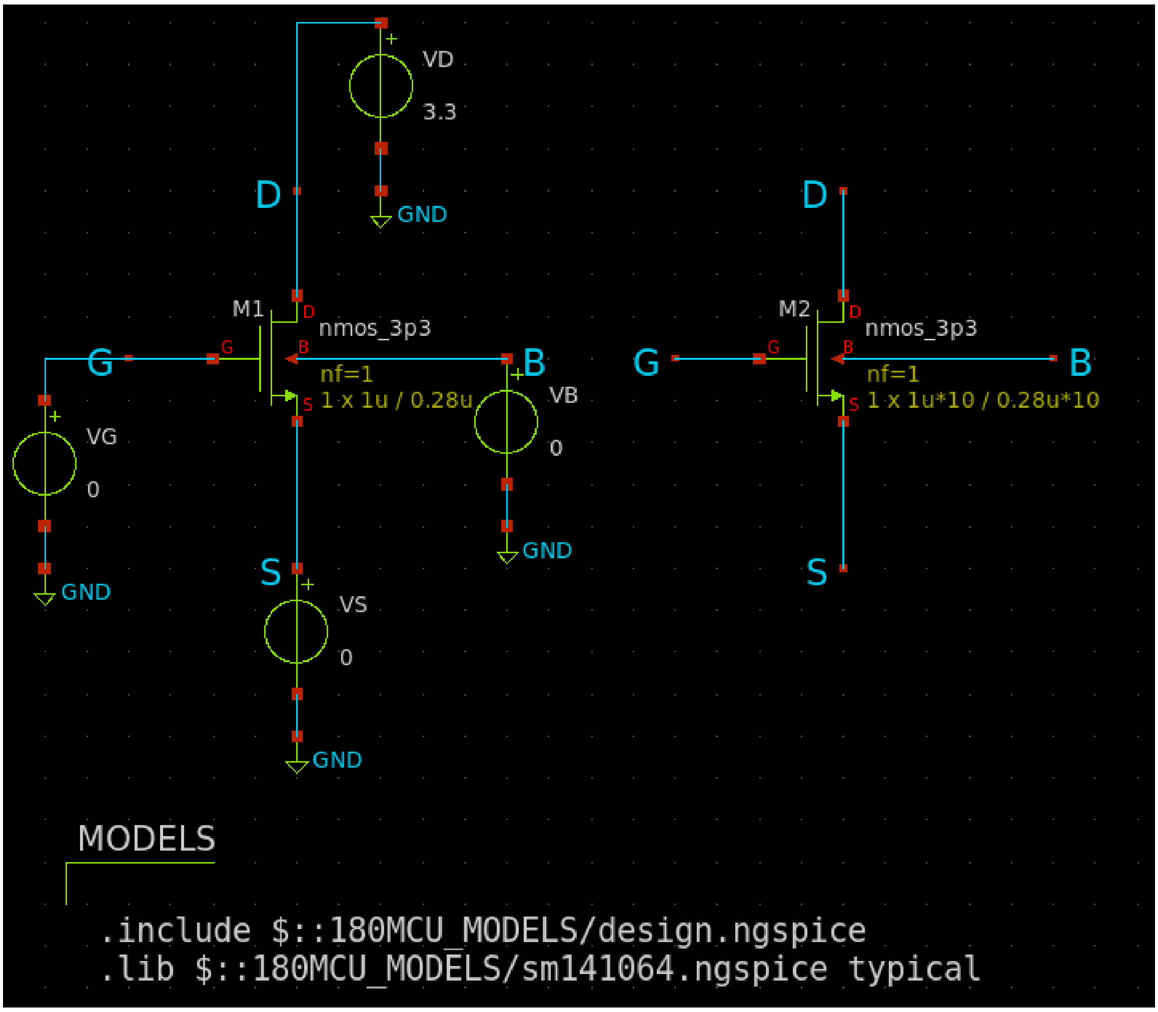
5) Modify the testbench to characterize NMOS and PMOS devices.
6) Plot $I_D - V_{GS}$ characteristics for NMOS and PMOS devices. Set $V_{DS} = V_{DD}$, and $V_{GS} = 0{:}\,10m{:}\,V_{DD}$ and $W = 10 * L$. Use $V_{DD} = 3.3V$ for GF180MCU technology. Plot the results overlaid for the following:
    - Short channel device: $L = 280nm$
    - Long channel device: $L = 2.8\mu m$
    - Hint: Note that W/L = 10 is the same for both cases.

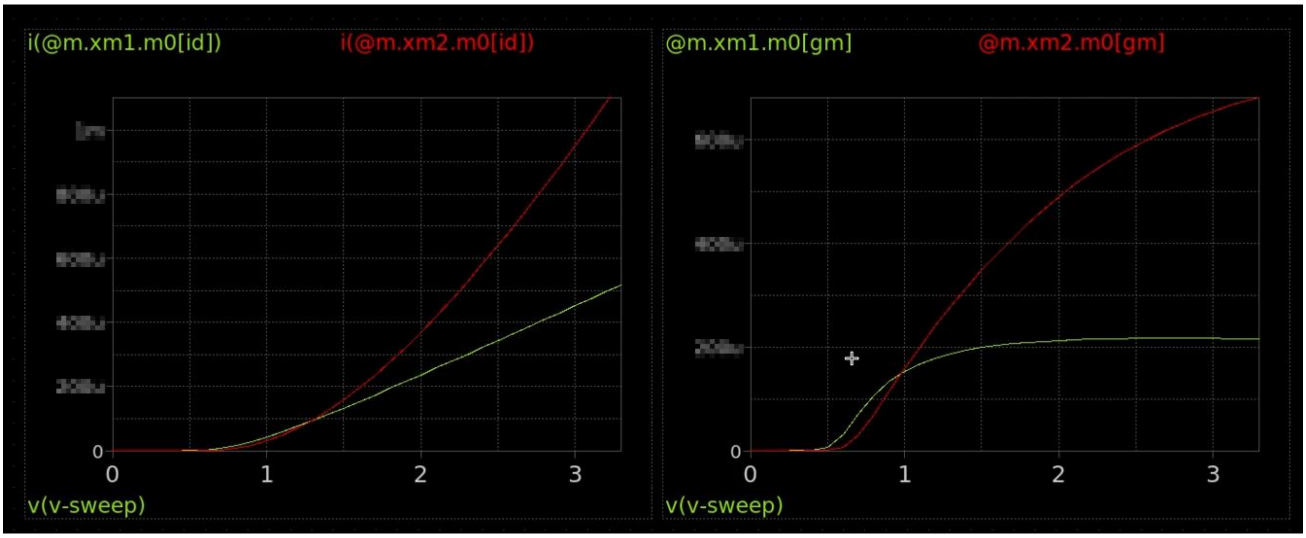
➔ Hint: Use DC sweep instead of parametric sweep whenever possible. DC sweep is extremely faster, and uses much less resources and disk space. In this question you should use DC sweep for VGS.

➔ Hint: To simulate both the short channel and the long channel devices in the same simulation run you can simply create two devices.

➔ Hint: This is an example of an NMOS short channel vs long channel testbench. Modify it as needed.

```
.control
save all
+ @m.xm1.m0[id] @m.xm1.m0[gm]
+ @m.xm2.m0[id] @m.xm2.m0[gm]
dc vg 0 3.3 0.1
*dc vd 0 2 0.01 vg 0 2 0.2
write test_mos.raw
.endc
```





2) Comment on the differences between short channel and long channel results.
   - Which one has higher current? Why?
   - Is the relation linear or quadratic? Why?
3) Comment on the differences between NMOS and PMOS.
   - Which one has higher current? Why?
   - What is the ratio between NMOS and PMOS currents at VGS = VDD?
   - Which one is more affected by short channel effects?

2. g_m vs VGS (use both Xschem and ADT)

1) Plot g_m vs V_{GS} for NMOS device. Set $W = 10 * L,\ V_{DS} = V_{DD}$, and $V_{GS} = 0{:}\,V_{DD}$. Plot the results overlaid for the following:
    - Short channel device: $L = 280nm$
    - Long channel device: $L = 2.8\mu m$
2) Comment on the differences between short channel and long channel results.
    - Does g_m increase linearly? Why?
    - Does g_m saturate? Why?

3. ID vs VDS (use ADT)

1) Plot $I_D - V_{DS}$ characteristics for NMOS device. Set $W = 10 * L, V_{DS} = 0{:}\,V_{DD}$ (primary sweep variable), and $V_{GS} = 1{:}\,1{:}\,3$ (secondary sweep variable of the nested sweep). Plot the results overlaid for the following:
    - Short channel device: $L = 280nm$
    - Long channel device: $L = 2.8\mu m$
2) Comment on the differences between short channel and long channel results.
    - Which one has higher current? Why?
    - Which one has higher slope in the saturation region? Why?

4. [Optional] g_m and r_o in Triode and Saturation (use ADT)

1) Plot g_m and $r_o\ vs\ V_{DS}$ for NMOS device. Use $W = 10 * L$ and $L = 2.8\mu m, V_{DS} = 0{:}\,V_{DD}$, and $V_{GS} \approx V_{TH} + 0.5V$.
   Hint: You can get an estimate of V_{TH} from the I_D vs V_{GS} characteristics you already plotted before, or you can print the operating point parameters of the transistor.
2) Comment on the variation of g_m vs V_{DS}.
    - In the first part of the curve, is the relation linear? Why?
    - Does g_m saturate? Why?
    - How much is V_{DS} when the curve saturates?
    - Where do you want to operate the transistor for analog amplifier applications? Why?
3) Comment on the variation of r_o vs V_{DS}.
    - Does r_o saturate just after the transistor enters saturation similar to gm? Why?
    - Does r_o increase if the transistor is biased more into saturation?
    - Should we operate the transistor at the edge of saturation?
    - Where do you want to operate the transistor for analog amplifier applications? Why?

Lab Summary

- In Part 1 you learned
    - How to run transient simulations.
    - How to run ac-analysis simulations.
    - How to run parametric sweeps.
- In Part 2 you learned
    - How to plot the transistors I/V characteristics using DC sweep.
    - How to plot the transistors characteristics using ADT.
    - The difference in transistor characteristics between an NMOS and a PMOS transistor.
    - The difference in transistor characteristics between a short-channel and a long-channel MOSFET.

- How the g_m of the transistor behaves vs VGS.
- How the g_m and r_o of the transistor behave in triode and in saturation.

Acknowledgements

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